

UltraSAM: A foundational medical ultrasound segmentation model with limited training data

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ABSTRACT

Segment Anything Models (SAMs) generalize well for generic vision tasks; however, they underperform on ultrasound images. Moreover, they require extensive annotated data for training. To address their limitations in generalization, high training data demands, and computational resource usage on ultrasound segmentation, we propose UltraSAM, a foundational medical ultrasound segmentation model. The main contributions of the study are as follows: First, we propose a Fast Fourier Transform (FFT) Convolutional (FFTConv) branch to enhance local frequency-domain feature extraction from ultrasound images, enabling better learning of generalizable ultrasound features. Second, we introduce an FFT injector (FFTI) to promote feature interaction between spatial and frequency domains, reducing training data requirements and improving the generalization of SAM on ultrasound images. Finally, low-rank adapters (LoRAs) are used to fine-tune the encoder, preserving the general visual representation of SAM while enabling domain adaptation for ultrasound imaging with minimal computational cost. We demonstrated the effectiveness of UltraSAM across four downstream segmentation tasks on 10 publicly available datasets spanning eight different ultrasound modalities. UltraSAM outperformed nine task-specific methods and six SAM-based foundation methods with only hundreds of annotated training samples. Compared with existing SAM-based segmentation methods, UltraSAM achieved a 2.07%–59.45% improvement in the Dice score in zero-shot segmentation tasks. In summary, UltraSAM enhances ultrasound diagnostics and personalized therapeutics by reducing dependence on resource-intensive datasets, holding potential to be a foundational clinical decision-support system.

1. Introduction

Ultrasound imaging has become clinically indispensable owing to its non-invasive, safe, and real-time diagnostic capabilities (Liu et al., 2019). However, traditional approaches to ultrasound image analysis face significant clinical application challenges. Image interpretation remains critically dependent on operator expertise, resulting in substantial inter-observer variability and a lack of standardized quantitative criteria. These limitations compromise diagnostic reliability and efficiency (Yin, Zheng, Chen, Hu, & Wang, 2025; Zihao Zhou et al., 2025). Recent advances in artificial intelligence, particularly in deep learning, have opened new avenues for automated ultrasound image analysis. Accurate automated delineation of anatomical structures and pathological lesions holds particular clinical value because it facilitates a

comprehensive quantitative morphological parameter assessment essential for diagnostic decision-making and treatment monitoring (Wang, Liu, Liu, & Chang, 2024).

The evolution of automated ultrasound image segmentation systems confronts three fundamental challenges rooted in prevailing task-specific design frameworks. First, current models demonstrate limited cross-domain adaptability, necessitating specialized architectural configurations for different anatomical regions or lesion subtypes. Second, each task typically requires the independent acquisition and annotation of large-scale datasets; this is resource-intensive and time-consuming. Third, compromised algorithmic robustness becomes evident through substantial efficacy reduction when deploying models in cross-device and cross-center applications (Wang et al., 2024). These issues critically constrain the clinical implementation of intelligent ultrasound

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analysis technologies.

Inspired by the success of foundation models in natural image, the medical imaging field is currently pursuing novel frameworks to develop cross-task generalizable models (Jiao et al., 2024; Ma et al., 2024). As a groundbreaking achievement in computer vision, a Segment Anything Model (SAM) demonstrates exceptional zero-shot generalization capability and adaptive prompt mechanisms, offering a new technical pathway for medical image segmentation tasks (Kirillov et al., 2023). Through its unified architecture design, the SAM integrates multi-task segmentation functionalities, significantly enhancing algorithmic deployment efficiency. This design provides critical insights for the development of general ultrasound image segmentation models capable of generalizing across different anatomical structures.

Nevertheless, inherent disparities between medical imaging modalities and natural image domains create substantial barriers for direct model adaptation (Ma et al., 2024). Three critical challenges emerge: First, ultrasound-specific characteristics—including speckle noise, acoustic artifacts, and tissue textural patterns—deviate markedly from natural image statistics. Second, current training paradigms insufficiently incorporate essential anatomical constraints and pathophysiological feature hierarchies. Third, vendor-specific image processing pipelines induce clinically significant inter-system domain shifts. These fundamental discrepancies ultimately limit the efficacy of foundation models in ultrasound analysis (Jiao et al., 2024).

Current methodologies predominantly employ transfer learning with domain adaptation techniques to develop specialized models via medical dataset fine-tuning of the SAM (Cheng et al., 2023; Ma et al., 2024; Wu et al., 2025; Zhang & Liu, 2023). Existing medical SAM extensions—such as MedSAM, SAM-Med2D, SAMUS, and MSA (Cheng et al., 2023; Lin et al., 2023; Ma et al., 2024; Wu et al., 2025)—are primarily trained on CT, MRI, and endoscopic images, with little or no inclusion of ultrasound data. Consequently, these models exhibit marked performance degradation when evaluated on public ultrasound benchmarks. This decline arises from two factors: (1) ultrasound-specific challenges absent from the training data, and (2) Current SAM implementations remain modality-agnostic and therefore neglect the distinctive physical properties inherent to ultrasound imaging. Hence, current medical SAM extensions cannot be directly applied to ultrasound data, motivating the ultrasound-tailored design of UltraSAM. The development of a universal ultrasound segmentation model faces the following obstacles: (1) Heavy reliance on annotated datasets conflicts with the specialized expertise requirements and prohibitive annotation costs of ultrasound; (2) Insufficient anatomical specificity exists in optimization strategies for the unique imaging physics of ultrasound. These constraints necessitate breakthroughs in three dimensions, namely (1) Scarcity of annotated data: Require advanced few-shot learning frameworks to mitigate data scarcity; (2) Inadequate cross-device generalization: Require solutions to domain adaptation issues caused by heterogeneous imaging parameters; (3) Weak cross-task generalization: Demand effective universal feature learning mechanisms to disentangle anatomical and pathological characteristics across organs.

This study presents UltraSAM, an optimized adaptation of the SAM specifically designed for ultrasound image segmentation in data-limited scenarios. To address high computational costs, large data requirements, and poor cross-modal transfer of existing SAM-based medical imaging approaches, we developed a novel framework that maintained the powerful feature representation of the SAM while overcoming these constraints. UltraSAM demonstrated robust performance across multiple organ and lesion segmentation tasks, achieving both computational efficiency and excellent domain generalization capabilities. The contributions of this study are threefold, directly addressing the key challenges of applying foundation models to ultrasound imaging under data-limited conditions:

1. This study proposes a novel foundational model, UltraSAM, that is designed for ultrasound image segmentation that excels with limited

data. By integrating Low-Rank adaptation (LoRA) (Hu et al., 2022), UltraSAM requires only hundreds of training samples and minimal computational resources, overcoming the prohibitive data and compute demands of traditional fine-tuning approaches

2. The proposed Fast Fourier transform (FFT) convolutional (FFTConv) branch and FFT injector (FFTI) enable hierarchical fusion of spatial and frequency-domain features, enhancing its generalization performance and robustness against domain shifts
3. We demonstrate superior cross-domain and zero-shot generalization. Trained on a small dataset of two lesion types, UltraSAM significantly outperforms nine task-specific methods and six other SAM-based models in segmenting six entirely unseen organ anatomies, proving its capability as a powerful, generalizable foundation for diverse clinical ultrasound applications

The remainder of this paper is organized as follows: Section 2 systematically reviews existing SAM-based approaches. Section 3 presents our technical framework. Section 4 presents experimental results, including both quantitative and qualitative analyses. The discussion in Section 5 critically analyzes the clinical implications and theoretical contributions. Section 6 concludes with key findings.

2. Related works

2.1. SAM

The effectiveness of deep learning in computer vision largely results from scaling neural network architectures supported by extensive training datasets. Current state-of-the-art image segmentation models predominantly utilize encoder networks that are pre-trained on large-scale datasets such as ImageNet and subsequently fine-tuned for specific tasks (Cao et al., 2022; Jiang et al., 2021; Ronneberger et al., 2015; Zongwei Zhou, Rahman Siddiquee, Tajbakhsh, & Liang, 2018). Although these approaches achieve strong performance on dedicated datasets, their zero-shot and few-shot generalization capabilities across varied scenarios remain substantially limited. The recent introduction of the SAM represents a breakthrough in universal image segmentation frameworks (Kirillov et al., 2023). Pre-trained on the SA-1B dataset—comprising 11 million natural images with one billion generated masks—the SAM demonstrates exceptional zero-shot segmentation performance for previously unseen targets. The innovative architecture of the model combines a prompt encoder with conventional encoder-decoder structures, enabling segmentation guidance through interactive prompts without requiring predefined semantic labels. This design facilitates precise instance object localization while eliminating the dependency on fixed semantic taxonomies that are characteristic of traditional methods, thereby establishing a new paradigm for open-world visual processing. This advancement highlights the considerable potential of foundation models in achieving cross-task generalization via large-scale pre-training, offering a viable approach to minimize redundant task-specific training requirements.

2.2. Fine-Tuning technologies in SAM

Although the SAM achieves remarkable zero-shot generalization in natural image segmentation, its direct application to medical imaging domains, particularly ultrasound segmentation, results in substantial performance degradation (Lin et al., 2023). Parameter-Efficient Fine-Tuning (PEFT) has emerged as a promising solution to bridge this domain gap while preserving computational efficiency. PEFT effectively mitigates catastrophic forgetting inherent to full-parameter fine-tuning while maintaining superior generalization in small-data and cross-domain scenarios (Wu et al., 2025). Within the context of medical image segmentation, Adapter (Wu et al., 2025) and LoRA (Hu et al., 2022) represent two predominant PEFT methodologies. Specifically, the Adapter method inserts lightweight supplementary modules between

pretrained layers to achieve task-specific feature adaptation with minimal computational cost. Conversely, LoRA injects low-rank matrix decompositions into model weights, simultaneously alleviating excessive dependence on the initial model and effectively suppressing overfitting. Both approaches enable performance optimization with minimal training cost, making them particularly suitable for data-scarce medical imaging applications. Thus, PEFT-based lightweight fine-tuning establishes the optimal pathway for transferring the SAM to ultrasound segmentation tasks.

2.3. SAM in medical image

Recent advances in medical image segmentation have demonstrated significant progress in adapting the SAM to clinical applications through three primary technical approaches. The first approach involves large-scale fine-tuning of general segmentation models. MedSAM (Ma et al., 2024) pioneered this approach by constructing a multimodal dataset containing 1.57 million medical image-mask pairs (spanning 10 imaging modalities and 30+ cancer types), coupled with prompt encoding mechanisms for full-parameter SAM fine-tuning, demonstrating the feasibility of foundation models in cross-modal medical segmentation. SAM-Med2D (J. Cheng et al., 2023) expanded training data to 4.6 million images and 19.7 million masks, achieving significant improvements in segmentation accuracy and generalization under multimodal prompts through joint adapter-based fine-tuning of the encoder and decoder of the SAM. The second technical direction focuses on PEFT to address computational constraints. Notable developments include the hierarchical decoding architecture of H-SAM that freezes the SAM encoder via LoRA while incorporating class-balanced attention mechanisms, reducing annotation requirements by 90%. Similarly, SAMed demonstrated that optimizing just 0.1% of parameters could achieve competitive performance of full fine-tuning performance, highlighting the efficiency of selective parameter updates.

Architectural adaptations constitute the third major approach, with innovations like SAMUS (Lin et al., 2023) introducing parallel CNN branches and cross-branch attention modules to enhance ultrasound segmentation. WeSAM (Zhang, Su, Xu, & Jia, 2024) further advanced this direction through anchor network regularization, enabling robust zero-shot adaptation to domain shifts without requiring source domain data. I-MedSAM (Wei et al., 2024) is designed for general medical images (e.g., CT and optics imaging), its Frequency Adapter (FA) primarily enhances high-frequency edge contours.

Our proposed UltraSAM addresses a fundamental gap in the field by introducing a novel architecture specifically tailored to the unique physical properties of ultrasound imaging. Unlike generic SAM variants, our model employs a domain-specific, hierarchical fusion mechanism—integrating frequency-domain cues via our proposed FFTConv branch and FFTI with spatial representations. This design effectively mitigates common ultrasound artifacts like speckle noise, enhances tissue characterization, and ensures robust generalization. This design contrasts with other strategies. For example, while another recently proposed model also named “Ultrasam” (Meyer, Murali, Zarin, Mutter, & Padoy, 2025) shares the goal of universality, it relies on training with large-scale open-access datasets. Our UltraSAM, however, is intentionally designed as a practical and powerful solution for the far more common clinical reality of data scarcity, achieving high-precision segmentation and superior cross-modal generalization with only hundreds of training samples. Fig. 1 illustrates the overall framework of the study. The left panel depicts the comprehensive medical imaging dataset, whereas the right panel presents an architectural comparison between the original SAM, existing medical-adapted SAM variants, and the proposed UltraSAM framework. Detailed explanations of the dataset construction and methodological innovations will be provided in subsequent sections.

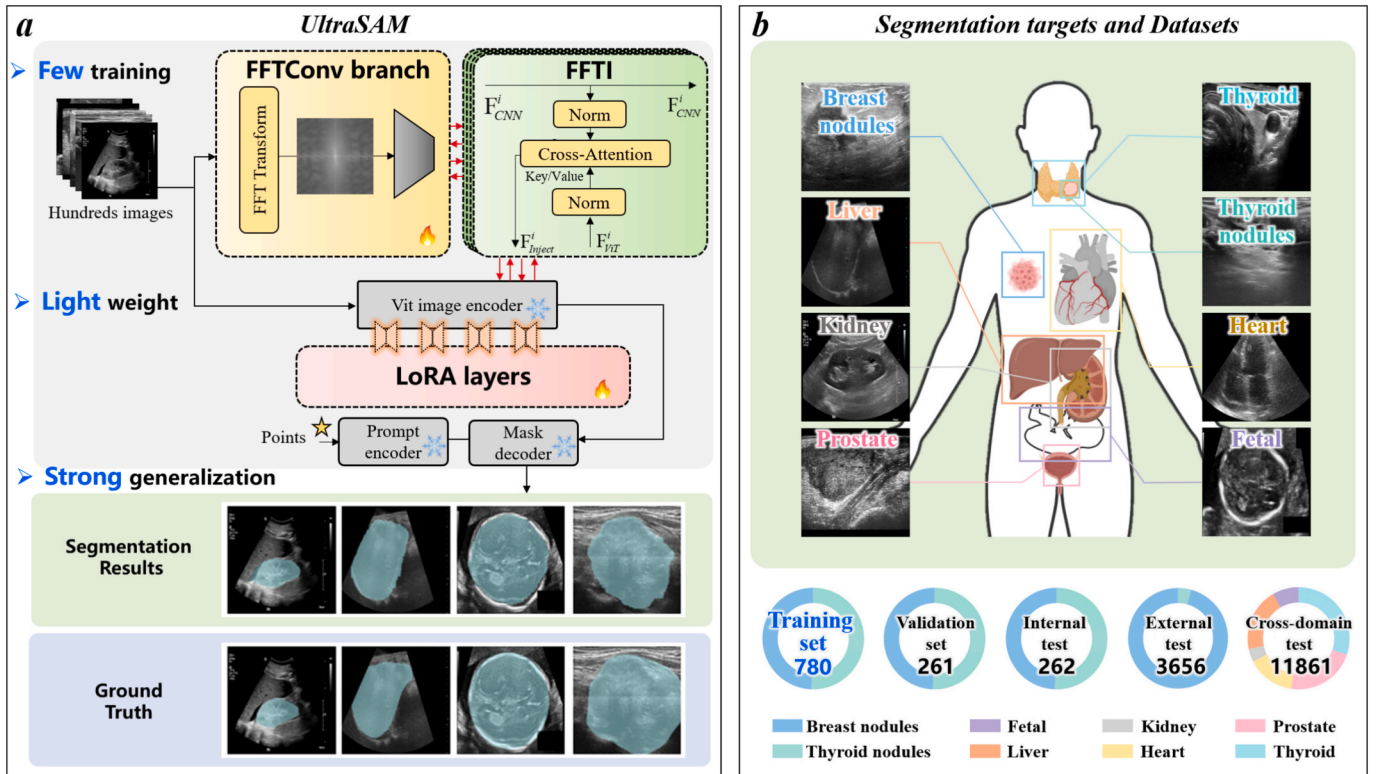


Fig. 1. Framework of the proposed methodology. (a) Proposed UltraSAM: Domain-specific adaptation is achieved via parameter-efficient partial fine-tuning using only hundreds of ultrasound images. The gray-shaded modules in the model represent the canonical SAM/Medical SAM architecture, which typically necessitates training on millions to billions of samples. (b) Data collection: Ultrasound images are acquired from two pathological categories and six anatomical organs.

3. Methods

3.1. Overview of the model

As shown in Fig. 2, UltraSAM builds on the core framework of the SAM by retaining the prompt encoder and mask decoder structures/parameters, while enhancing the encoder through the following three key modifications. (1) FFTConv branch: FFTConv operates directly in the frequency domain, exploiting the physics of ultrasound echoes to suppress speckle noise, capture subtle spectral shifts from small lesions, and yielding a more robust, detail-sensitive, and device-agnostic features. (2) FFTI: It fuses FFT features from the CNN branch with global spatial-domain features from the ViT branch through cross-modal interaction, achieving complementary spatial-frequency feature enhancement. At each layer, the FFTI intelligently combines frequency features (from FFTConv) with spatial features (from ViT), helping the model decide whether to rely more on spectral or spatial information. This design effectively addresses feature loss of small targets caused by patch partitioning by ViT and reduces spectral interference from ultrasound artifacts. (3) LoRA: PEFT is achieved via low-rank matrix decomposition, enabling ultrasound domain adaptation while preserving the general visual representation capabilities of the SAM. Detailed technical specifications of each module are provided in subsequent sections.

3.2. Fftconv branch

The FFTConv branch is designed to leverage the fundamental properties of ultrasound imaging by operating directly in the frequency domain. Since ultrasound B-mode images are reconstructed from radio-frequency signals via Fourier transform, frequency content provides a physically-grounded representation of tissue properties (Hasegawa, 2021; Vahid, Prabhu, Farhadi, & Rastegari, 2020). This approach offers three key advantages: (1) mitigating characteristic speckle noise patterns; (2) detecting subtle spectral shifts from low-contrast lesions; and (3) normalizing scanner-specific variations for better cross-device generalization. By processing frequency spectra directly, FFTConv effectively utilizes these intrinsic ultrasound properties. The FFT (Wei et al., 2024), transforms an original image $f(h, w) \in \mathbb{R}^{H \times W}$ into a complex spectral representation as follows:

$$F(u, v) = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} f(h, w) \cdot e^{-j2\pi \left(\frac{hu}{H} + \frac{vw}{W} \right)} \quad (u \in [0, H], v \in [0, W]) \quad (1)$$

where the magnitude spectrum, $|F(u, v)|$, characterizes frequency-domain energy distribution, and the phase spectrum, $\arg(F(u, v))$, encodes spatial structural information. As illustrated in Fig. 3, the input image, $\mathbf{X}_{in}$, is first subjected to FFT transformation as follows:

$$\mathbf{X}_{FFT} = FFT(\mathbf{X}_{in}) \quad (2)$$

Subsequently, a CNN branch is constructed to process frequency-domain features. This branch comprises an initial convolutional block (Convolution–Batch Normalization–ReLU) followed by three down-sampling stages. Each stage combines a 3×3 convolutional layer with max-pooling (stride = 2). Through progressive feature extraction, frequency-domain feature maps $\mathbf{F}_{FFT} \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C}$ aligned with the spatial resolution of the ViT branch are ultimately generated, achieving spatial-frequency dual-domain feature alignment at identical scales.

3.3. FFTI

An FFTI is proposed to integrate frequency-domain prior information from FFT into the ViT encoder. Compared with prior works that apply FFT as a static preprocessing step, FFTConv extracts hierarchical frequency features through a trainable CNN branch. Coupled with FFTI's cross-attention mechanism, this enables adaptive recalibration of spatial features using frequency priors. In the i -th encoder block of ViT, the input visual feature tensor, $F_i^{ViT} \in \mathbb{R}^{N \times d}$ (where N is the number of image patches and d is the feature dimension), serves as the query in the cross-attention mechanism. The frequency-domain features extracted by FFT, $F_i^{fft} \in \mathbb{R}^{M \times d}$, are linearly projected into key and value vectors. This allows spatial-frequency information from FFT features to be dynamically injected into the ViT backbone through a learnable cross-attention mechanism. The process begins with layer normalization of the input features as follows:

$$\hat{F}_i^{vit} = F_i^{vit} + \text{Attention} \left(\text{LayerNorm}(F_i^{vit}), \text{LayerNorm}(F_i^{fft}), \text{LayerNorm}(F_i^{fft}) \right) \quad (3)$$

The cross-attention mechanism is then implemented as follows:

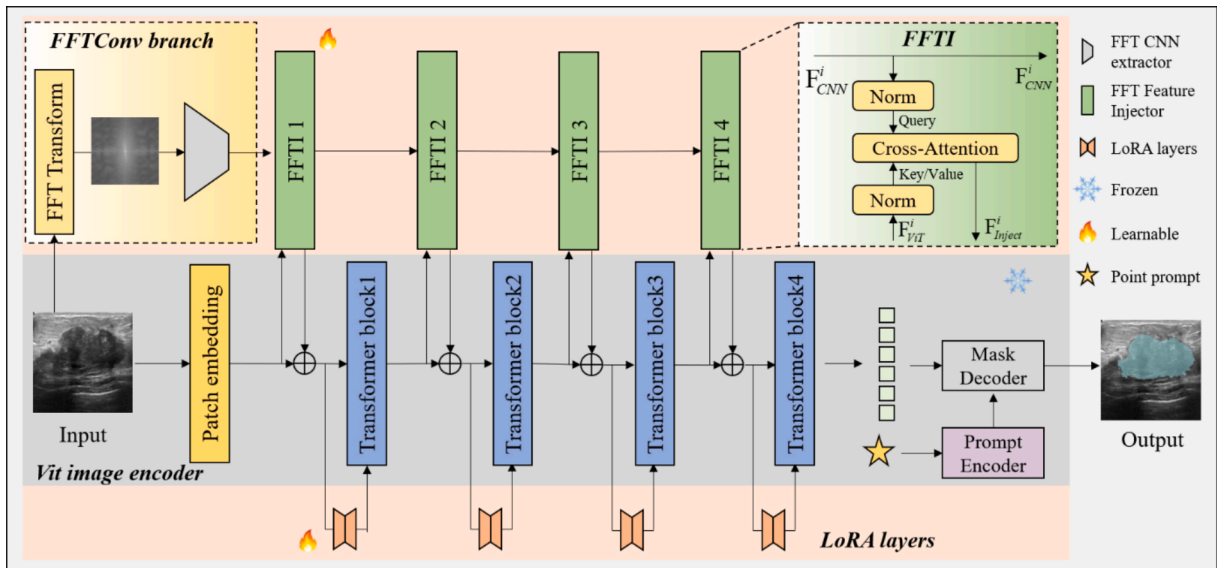


Fig. 2. Overview of the UltraSAM model architecture and methodology.

Algorithm 1 UltraSAM: Ultrasound-Specific Segmentation with FFTConv, FFT Injector and LoRA Adaptation

Require: Input ultrasound image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$
Require: Prompt point p
Require: Pre-trained SAM ViT-B backbone $\mathcal{S}_0$
Require: FFTConv branch $\mathcal{F}(\cdot)$
Require: FFT injector $\mathcal{I}(\cdot, \cdot)$
Require: LoRA adapters $\mathcal{L}_Q, \mathcal{L}_V$ on ViT Query & Value
Require: Binary segmentation mask $\mathbf{M} \in \{0, 1\}^{H \times W}$

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1: function BUILDULTRASAM( $\mathcal{S}_0$ )
2:   Freeze all parameters of  $\mathcal{S}_0$  except LoRA adapters  $\mathcal{L}_Q, \mathcal{L}_V$ 
3:   Insert  $\mathcal{F}$  parallel to ViT block
4:   Insert  $\mathcal{I}$  into each ViT block
5:   return model  $\mathcal{U} = \{\mathcal{S}_0, \mathcal{F}, \mathcal{I}, \mathcal{L}_Q, \mathcal{L}_V\}$ 
6: end function
7: function FORWARD( $\mathcal{U}, \mathbf{I}, p$ )
8:    $\mathbf{F} \leftarrow \mathcal{F}(\mathbf{I})$ 
9:    $\mathbf{T} \leftarrow \text{PatchEmbed}(\mathbf{I})$ 
10:  for each ViT block  $l$  do
11:     $\mathbf{T} \leftarrow \mathcal{I}_l(\mathbf{T}, \mathbf{F})$ 
12:     $\mathbf{T} \leftarrow \text{ViTBlock}_l(\mathbf{T})$ 
13:  end for
14:   $\mathbf{E} \leftarrow \text{Neck}(\mathbf{T})$ 
15:   $\mathbf{M} \leftarrow \text{MaskDecoder}(\mathbf{E}, p)$ 
16:  return  $\mathbf{M}$ 
17: end function
18: function TRAINULTRASAM
19:   $\mathcal{U} \leftarrow \text{BUILDULTRASAM}(\mathcal{S}_0)$ 
20:  optimizer  $\leftarrow \text{AdamW}(\{\mathcal{F}, \mathcal{I}, \mathcal{L}_Q, \mathcal{L}_V\}, \text{lr} = 1\text{e-}4)$ 
21:  loss  $\leftarrow \text{DiceCELoss}()$ 
22:  for epoch = 1 to  $E$  do
23:    for ( $\mathbf{I}, \text{GT}, p$ ) in dataloader do
24:      Pred  $\leftarrow \text{FORWARD}(\mathcal{U}, \mathbf{I}, p)$ 
25:       $\ell \leftarrow \text{loss}(\text{Pred}, \text{GT})$ 
26:      Backward( $\ell$ ); Step(optimizer)
27:    end for
28:  end for
29:  return  $\mathcal{U}$ 
30: end function

```

Fig. 3. Pseudocode of UltraSAM.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where d_k is the dimension of the key vectors.

This architecture enables adaptive adjustment of attention distributions across image regions based on spatial correlations derived from frequency-domain features. The layer-wise injection strategy progressively enhances the perception of texture details and contour features.

3.4. LoRA

UltraSAM strategically retains SAM's generic knowledge by integrating LoRA (Hu et al., 2022), thereby achieving effective ultrasound-domain adaptation while preventing overfitting on small-scale datasets. Given an input token sequence $F \in \mathbb{R}^{B \times N \times C_{in}}$, the frozen projection layer, $W \in \mathbb{R}^{C_{out} \times C_{in}}$, linearly transforms it into an output token sequence $F' = WF$. LoRA efficiently updates parameters via low-rank decomposition that can be formulated as follows:

$$W^o = W + \Delta W = W + BA \quad (5)$$

where $B \in \mathbb{R}^{C_{out} \times r}$ and $A \in \mathbb{R}^{r \times C_{in}}$ form a low-rank matrix pair ($r \ll \min(C_{in}, C_{out})$), allowing the weight matrix to be adjusted by training only a small fraction of parameters. Leveraging the multi-head attention mechanism features, LoRA adapters are selectively inserted into the query (Q) and value (V) projection layers. This maintains the original spatial attention patterns of the SAM and adjusts attention distributions via trainable parameters. The implementation can be formulated as follows:

$$\begin{cases} Q = \omega_q F = (\omega_q + \beta_q \alpha_q) F \\ K = \omega_k F \\ V = \omega_v F = (\omega_v + \beta_v \alpha_v) F \end{cases} \quad (6)$$

Here, ω_q, ω_k , and ω_v denote frozen original projection matrices, and $\{\alpha_q, \beta_q, \alpha_v, \beta_v\}$ denote trainable parameters. Experiments showed that this setup greatly boosted ultrasound image segmentation performance

while maintaining model stability.

3.5. Training strategies

The UltraSAM framework initializes the SAM encoder with a ViT-B backbone pretrained on the SA-1B dataset, whereas the other parameters are randomly initialized. During training, only the parameters from the FFTI and LoRA modules are updated, and the remaining parameters are frozen. The LoRA rank is set to 4. A combined loss function of Dice loss and binary cross-entropy loss is used for optimization. Consistent with SAM and other interactive segmentation frameworks (Ma et al., 2024; Wu et al., 2025; Cheng et al., 2023; Lin et al., 2023), UltraSAM is trained by emulating interactive segmentation. It employs a single positive point prompt, obtained by randomly sampling one pixel from the expert-annotated foreground region to approximate clinical annotation behavior. The Adam optimizer with a momentum of 0.9, an initial learning rate of $1e-4$, and a batch size of 8 is used. All comparative methods are trained for 400 epochs under the same settings. A pseudocode is supplemented to clearly elucidate the implementation of our proposed method (Fig. 3).

4. Results

4.1. Datasets

To comprehensively evaluate UltraSAM, a large-scale ultrasound dataset was constructed, spanning multi-organs, multi-centers, and multi-vendors. This dataset included two pathological categories and six anatomical structures, with 16,820 cases from 10 medical centers and major ultrasound vendors. The dataset construction adhered to the following protocols:

1. Task-specific training/internal test sets: BUSI (Al-Dhabyani, Goma, Khaled, & Fahmy, 2020) (breast nodules) and DDTI (Pedraza et al., 2015) (thyroid nodules) datasets were split via stratified sampling based on benign/malignant subtypes in a 6:2:2 ratio for training, validation, and internal testing.
2. Task-agnostic training/internal test sets: A Mix dataset (BUSI + DDTI) was created for foundational model training and internal evaluation of general segmentation capabilities
3. External test sets: Independently acquired Dataset B (breast nodules) (Yap et al., 2017) and TN3k (Gong et al., 2021) (thyroid nodules) dataset were utilized for cross-center performance assessment.
4. Zero-shot generalization test sets: Six specialized benchmarks were included: Infant head (HC) (van den Heuvel et al., 2018), heart (CAMUS) (Lecierc et al., 2019), liver (CLUST) (Ansari, Yang, Meher, & Dakua, 2023), kidney (Kidney-US) (Singla et al., 2023), prostate (Micro-US) (Hongxu Jiang et al., 2024), and thyroid (TG3k) (Gong et al., 2021), enabling rigorous evaluation of unseen organ segmentation under diverse imaging protocols. Details are presented in Table 1

Table 1
Dataset characteristics.

Task	Target	Dataset	Train	Validation	Test	Public
Task-specific training & internal test	Breast nodule	BUSI	388	129	130	Yes
	Thyroid nodule	DDTI	392	132	132	Yes
Task-agnostic training & internal test	Breast nodule & Thyroid nodule	Mix	780	261	262	Yes
	Breast nodule	Dataset B	—	—	163	Yes
	Thyroid nodule	TN3k	—	—	3493	Yes
Zero-shot generalization test	Fetal head	HC	—	—	995	Yes
	Liver	CLUST	—	—	2372	Yes
	Heart	CAMUS	—	—	1800	Yes
	Kidney	Kidney-US	—	—	488	Yes
	Prostate	Micro-US	—	—	2621	Yes
	Thyroid	TG3k	—	—	3585	Yes

4.2. Comparison methods

To systematically evaluate the segmentation performance and generalization capacity of UltraSAM across diverse clinical scenarios, we established a multi-tier validation framework comprising four experimental protocols, namely 1) Intra-dataset validation assessing fundamental capabilities through homogeneous training and evaluation on BUSI, DDTI, and Mix datasets; 2) Cross-domain generalization testing robustness against acquisition variations using external datasets (Dataset B and TN3k) from identical anatomical sources; 3) Cross-modality zero-shot validation evaluating unsupervised adaptation through direct application to six distinct anatomical regions (HC, CLUST, CAMUS, Kidney-US, Micro-US, and TG3k) acquired under different imaging protocols; 4) Pre-training comparative analysis quantifying data efficiency advantages by benchmarking against foundation models pre-trained on million-scale medical image corpora.

Comparative methods encompass three categories, namely classic CNN implementations including ResUNet (Diakogiannis, Waldner, Caccetta, & Wu, 2020), UNet++ (Zongwei Zhou et al., 2018), DeepLabv3+ (L.-C. Chen, Zhu, Papandreou, Schroff, & Adam, 2018), and UperNet (Xiao, Liu, Zhou, Jiang, & Sun, 2018); ViT models including TransUNet (Jiang et al., 2021), SwinUNet (Cao et al., 2022), TransFuse (Zhang et al., 2021), and MissFormer (Huang et al., 2021); and SAM-based foundation models including SAM, SAMed (Zhang & Liu, 2023), SAM-Med2D (Cheng et al., 2023), MSA (Wu et al., 2025), H-SAM (Cheng et al., 2024), and SAMUS (Lin et al., 2023). Following the same protocol as SAM, SAM-Med2D, MSA, and SAMUS et al, we feed the prompt encoder with automatically simulated point prompts, a procedure that aligns with medical-SAM evaluation standards, preserves clinical relevance, and eliminates human bias. Quantitative performance assessment employed Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) as primary metrics (Liu, Dong, Zhou, Yao, & Liu, 2025).

4.3. Ablation study

To quantify the individual contributions of UltraSAM, we conducted ablation studies across the BUSI, DDTI, and Mix datasets under consistent hyperparameter settings with five-fold cross-validation. Our analysis progressively evaluated four key components, namely CNN branch, FFT-enhanced CNN pathway (FFTConv), FFTI, and LoRA, reporting mean Dice/IoU metrics with standard deviations. The results are summarized in Table 2.

The ablation results demonstrated incremental and statistically significant improvements in performance as each component was added to the baseline model. The baseline model achieved Dice scores of $83.66 \pm 5.40\%$, $85.16 \pm 3.01\%$, and $72.61 \pm 4.28\%$ on the BUSI, DDTI, and Mix datasets, respectively. Integrating the CNN branch led to a notable improvement on the Mix dataset, with a 7.77% increase in Dice and a 7.51% increase in IoU. The incorporation of FFTConv further significantly enhanced model robustness across all datasets ($p < 0.01$), particularly on the Mix dataset, where the Dice score improved to 85.55

Table 2

Ablation study on different component combinations of UltraSAM on the BUSI, DDTI, and Mix datasets.

Components				BUSI		DDTI		Mix	
CNN	FFT	FFTI	LoRA	Dice	IoU	Dice	IoU	Dice	IoU
0	0	0	0	83.66 ± 5.40	74.43 ± 6.41	85.16 ± 3.01	75.98 ± 3.16	72.61 ± 4.28	62.52 ± 3.91
1	0	0	0	83.82 ± 6.28	74.85 ± 7.39	86.01 ± 3.66*	77.17 ± 4.40*	80.38 ± 0.96*	70.03 ± 0.97*
1	1	0	0	85.78 ± 4.40*	77.15 ± 5.35*	87.02 ± 2.87*	78.44 ± 3.43*	85.55 ± 2.97*	76.55 ± 3.54*
1	1	1	0	86.71 ± 5.18*	78.61 ± 6.10*	88.01 ± 3.45*	80.16 ± 4.41*	86.62 ± 3.08*	78.58 ± 3.87*
1	1	1	1	87.90 ± 4.68*	80.21 ± 5.69*	89.76 ± 2.89	82.17 ± 4.19*	86.80 ± 3.25	78.60 ± 3.91

* At the 0.01 level, the segmentation performance is statistically significantly different from the previous component combinations.

± 2.97 %, representing a 12.94 % total gain over the baseline. The subsequent addition of the FFTI module provided further task-specific and statistically significant enhancements ($p < 0.01$), with Dice scores increased to 86.71 ± 5.18 % on BUSI and 88.01 ± 3.45 % on DDTI.

The complete model architecture, including LoRA adaptation, achieved the highest performance across all datasets: 87.90 ± 4.68 % Dice (+4.24 %) on the BUSI, 89.76 ± 2.89 % Dice (+4.60 %) on the DDTI, and 86.80 ± 3.25 % Dice (+14.19 %) on the Mix datasets. LoRA fine-tuning delivered significant improvements for BUSI (+1.19 % Dice, $p < 0.01$) and DDTI IoU (+2.01 %, $p < 0.01$), although its effect on the Mix dataset Dice was not statistically significant. Additional experiments evaluating the impact of LoRA rank indicated that a rank of 4 provided the optimal balance between parameter efficiency and model effectiveness.

4.4. Comparison with SOTA Task-Specific methods on the internal test set

This section provides an assessment of the UltraSAM segmentation performance on breast and thyroid tumors via separate training and validation on the BUSI, DDTI, and Mix datasets, with comparisons to task-specific methods. As presented in Table 3 and Fig. 4, when trained on a single dataset (BUSI or DDTI), UltraSAM delivered optimal performance on the homologous test sets. Specifically, training and testing on the BUSI dataset yielded a Dice score of 80.22 % and an IoU score of 71.11 %. Similarly, training and testing on the DDTI dataset resulted in a Dice score of 84.83 % and an IoU score of 75.08 %. In contrast, the CNN model demonstrated a stable performance within the same domain, achieving an average Dice score of 73.10 % on the BUSI dataset. However, its performance significantly degraded in cross-domain scenarios, with the average Dice score dropping to 31.10 % when the model trained on the BUSI dataset was tested on the DDTI dataset.

The Transformer model exhibited better cross-domain adaptability, achieving an average Dice score of 39.30 % when trained on the DDTI dataset and tested on the BUSI dataset. Yet, it still showed considerable performance fluctuations. However, UltraSAM achieved a Dice score of 62.06 % when trained on the BUSI dataset and tested on the DDTI dataset, marking a 22.78 % improvement over the best comparative method (UNet++ that scored 39.28 %). This result underscored the

UltraSAM capacity to effectively capture common features in ultrasound images through few-shot learning. Moreover, when trained on the Mix dataset, UltraSAM showed remarkable cross-domain generalization. The model trained on the Mix dataset and tested on the BUSI dataset attained a Dice score of 79.85 %; this was only a 0.37 % decrease compared with the Dice score of the model trained and tested only on the BUSI dataset. Even more impressively, when tested on the DDTI dataset, the Mix-trained model achieved a Dice score of 85.15 %, a 0.32 % increase compared with the Dice score of the model trained and tested on the DDTI dataset. Collectively, these findings confirmed that UltraSAM, leveraging frequency-domain feature optimization and dynamic prompting mechanisms, achieved superior performance and generalizability in ultrasound image segmentation, particularly when trained on the Mix dataset.

Fig. 5 presents the qualitative analysis. During training and testing on the same dataset, the proposed method achieved more accurate segmentation of the target region, with results closely aligning with the ground truth. Other methods, such as ResUNet, UNet++, and DeepLabv3+, exhibited over-segmentation or under-segmentation in certain regions. In cross-task segmentation scenarios, UltraSAM maintained a high level of segmentation accuracy, whereas other task-specific methods failed to produce meaningful segmentation results. When trained on the Mix dataset, the proposed method demonstrated the ability to precisely segment both breast and thyroid nodules simultaneously, confirming that the mixed training effectively adapted to diverse image characteristics.

4.5. Comparison with SOTA foundation models on the internal test set

We further compared UltraSAM with SAM-based baseline models on the internal test set. As presented in Table 4 and Fig. 4, when trained on a single dataset, UltraSAM outperformed the second-best model, SAMUS (BUSI: 79.82 %; DDTI: 83.78 %), by 0.394 % on the BUSI dataset and by 1.051 % on the DDTI dataset. SAMed delivered a stable performance in single-organ tasks through LoRA, achieving a Dice score of 83.91 % and an IoU score of 74.02 % when trained and tested on the DDTI dataset. However, its cross-domain generalization ability was significantly outperformed by SAMUS that incorporated parallel CNN branches (BUSI

Table 3

Quantitative comparison of UltraSAM and SOTA task-specific methods on the internal test set. The best results are marked in bold.

Training set	BUSI				DDTI				Mix			
	BUSI		DDTI		BUSI		DDTI		BUSI		DDTI	
Test set												
Methods	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
ResUNet	74.80	61.25	21.65	13.21	37.57	24.46	84.01	72.89	73.93	60.40	84.50	73.52
UNet++	71.11	56.80	39.28	25.11	08.75	15.19	83.70	72.38	75.00	61.48	84.42	74.85
DeepLabv3+	73.70	60.37	25.03	15.48	39.82	25.66	83.85	72.64	70.86	56.49	84.39	73.30
UperNet	72.62	58.60	38.81	25.04	46.51	30.78	82.43	70.42	73.31	59.52	83.03	71.27
SegFormer	73.99	60.28	13.74	07.99	40.44	26.11	82.21	70.14	74.97	61.34	83.16	71.61
TransUNet	68.41	53.24	38.67	24.26	38.67	24.26	74.28	59.69	69.10	54.60	74.02	59.61
SwinUNet	56.99	40.89	29.86	18.00	38.78	24.38	66.37	50.08	53.94	37.58	64.37	47.93
TransFuse	71.53	57.17	25.88	15.27	41.41	26.50	77.93	64.49	68.72	53.70	79.17	66.06
MissFormer	65.84	50.48	24.01	14.05	37.56	23.55	72.20	56.87	61.80	45.69	67.34	51.30
UltraSAM	80.22	71.11	62.06	49.58	63.75	53.37	84.83	75.08	79.85	70.80	85.15	75.72

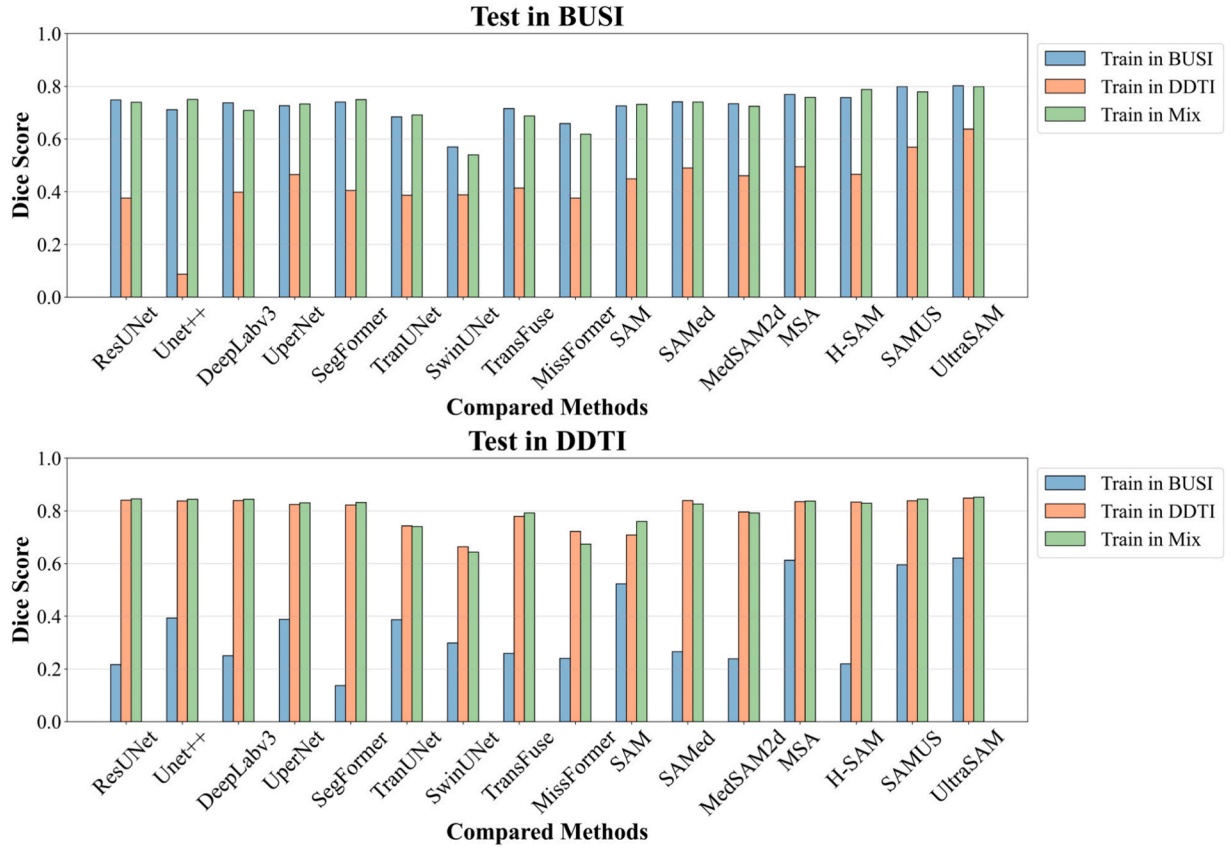


Fig. 4. Comparison between UltraSAM and task-specific and foundation methods evaluated on the BUSI (top) and DDTI (bottom) internal test sets.

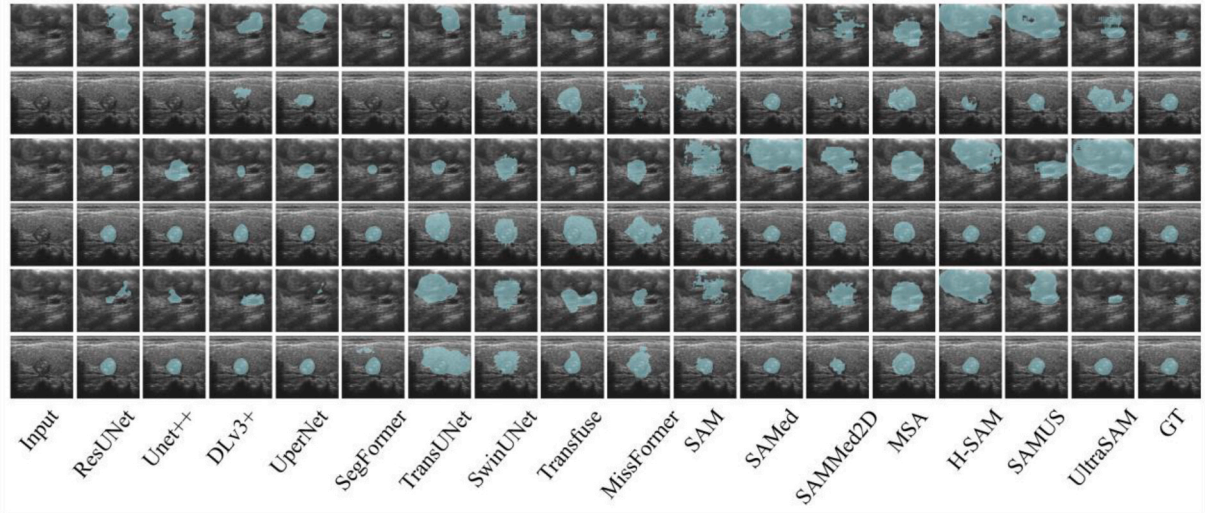


Fig. 5. Qualitative comparison between UltraSAM and task-specific and foundation methods on the internal test set. Row 1: training on BUSI and testing on BUSI (BUSI-BUSI); Row 2: BUSI-DDTI; Row 3: DDTI-BUSI; Row 4: DDTI-DDTI; Row 5: Mix-BUSI; Row 6: Mix-DDTI.

trained and DDTI tested; Dice: 26.55 %; IoU: 19.75 %), underscoring the importance of local feature supplementation for ultrasound image adaptability. H-SAM enhanced task-specific accuracy via hierarchical decoding when trained and tested on DDTI (Dice: 83.26 %; IoU: 73.61 %). Yet, its cross-domain performance was less than 30 % of that of UltraSAM owing to the absence of a cross-modal adaptation mechanism (BUSI trained and DDTI tested; Dice: 21.94 %; IoU: 16.61 %). In cross-domain zero-shot tasks (BUSI trained and DDTI tested), the UltraSAM Dice score was 2.533 % higher than that of SAMUS (59.53 %). Similarly,

in the reverse task (DDTI trained and BUSI tested), it outperformed the second-best model SAMUS (56.923 %) by 6.829 %. When trained on the Mix dataset, the cross-organ generalization ability of UltraSAM significantly exceeded that of other models, confirming its superior capability in fusing multi-organ features.

Fig. 5 presents the qualitative analysis. Other SAM-based models are prone to noise interference in complex scenes, leading to inaccurate segmentation results. Compared with other SAM-based models, the proposed UltraSAM demonstrated superior performance in terms of

Table 4

Quantitative comparison of UltraSAM and foundation models on the internal test set. The best results are marked in bold.

Training set	BUSI				DDTI				Mix			
Test set	BUSI		DDTI		BUSI		DDTI		BUSI		DDTI	
Methods	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
SAM	72.54	61.03	52.32	37.33	44.85	33.11	70.80	56.90	73.13	61.56	75.95	62.32
SAMed	74.09	65.25	26.55	19.75	48.96	39.33	83.91	74.01	73.99	65.52	82.60	72.60
SAM-Med2D	73.33	64.83	23.90	15.31	46.12	37.11	79.54	75.01	72.41	60.25	79.17	70.46
MSA	76.85	66.37	61.22	45.46	49.47	40.36	83.44	73.43	75.83	66.31	83.68	73.89
H-SAM	75.71	67.01	21.94	16.61	46.57	37.86	83.26	73.61	78.78	69.98	82.83	72.96
SAMUS	79.82	70.74	59.53	46.41	56.92	46.09	83.78	73.76	77.92	68.50	84.47	74.44
UltraSAM	80.22	71.11	62.06	49.58	63.75	53.37	84.83	75.08	79.85	70.80	85.15	75.72

segmentation accuracy, detail capture, noise robustness, and generalization ability. This was particularly evident in its outstanding performance on the Mix dataset training.

4.6. Comparison with SOTA task-specific methods on the external test set

We evaluated the cross-center generalization capabilities of UltraSAM by training the model on single-organ datasets (BUSI, DDTI, and Mix) and testing it on cross-center datasets (Dataset B and TN3k). As shown in Table 5 and Fig. 6, the experimental results demonstrated that UltraSAM achieved superior performance than task-specific baseline models in cross-center ultrasound segmentation tasks. When trained on single-organ datasets, UltraSAM exhibited consistent superiority in homologous cross-center testing. Specifically, training on the BUSI dataset and testing on Dataset B yielded a Dice score of 79.69 % and an IoU score of 70.82 %. Similarly, training on the DDTI dataset and testing on the TN3k dataset achieved a Dice score of 67.23 % and an IoU score of 56.45 %. In contrast, CNN models demonstrated significant performance degradation in cross-task cross-center testing scenarios. For instance, when trained on the BUSI dataset and tested on the TN3k dataset, the average Dice score was 52.55 %, whereas training on the DDTI dataset and testing on Dataset B resulted in an average Dice score of 27.18 %.

However, UltraSAM demonstrated exceptional resilience in these challenging scenarios. When trained on the BUSI dataset and tested on the TN3k dataset, UltraSAM achieved a Dice score of 67.19 %, surpassing the best-performing baseline model Deeplabv3+ by 12.3 % (Deeplabv3+ Dice score: 54.89 %). In the DDTI-to-BUS testing scenario, UltraSAM attained a Dice score of 47.55 %, outperforming all baseline models by a significant margin, including a 15.18 % improvement over the suboptimal model SegFormer (SegFormer Dice score: 32.37 %). UltraSAM further demonstrated exceptional cross-organ and cross-center generalization capabilities when trained on the Mix dataset. In the Mix-to-TN3k testing scenario, UltraSAM achieved a Dice score of 73.38 %, representing a 6.15 % improvement over the DDTI-to-TN3k testing scenario. These results underscored the robustness of UltraSAM in handling diverse testing conditions across datasets and tasks.

Table 5

Quantitative comparison of UltraSAM and SOTA task-specific methods on the external test set. The best results are marked in bold.

Training set	BUSI				DDTI				Mix			
Test set	Dataset B		TN3k		Dataset B		TN3k		Dataset B		TN3k	
Methods	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
ResUNet	69.33	55.04	53.76	38.17	29.25	18.02	61.05	45.30	61.81	47.11	64.99	49.44
UNet++	70.69	57.72	47.79	32.77	25.32	15.17	60.34	44.72	58.68	43.87	67.17	51.94
DeepLabv3+	68.99	54.34	54.89	39.17	23.96	14.06	60.15	44.39	60.84	45.43	64.68	49.03
UperNet	62.48	47.78	53.74	37.95	30.20	18.79	61.93	46.12	56.36	41.40	66.15	50.67
SegFormer	67.99	53.41	54.50	38.85	32.37	20.38	60.58	44.68	59.67	44.55	65.97	50.54
TransUNet	48.95	34.47	50.09	34.61	21.39	12.39	47.58	31.85	44.33	30.46	61.28	45.51
SwinUNet	45.35	31.06	40.89	26.71	17.56	09.85	43.34	28.33	40.85	26.88	48.18	32.64
TransFuse	44.89	30.57	45.46	30.65	24.43	14.54	50.67	35.03	43.15	29.08	58.92	43.10
MissFormer	58.00	42.87	43.38	28.61	19.87	11.44	43.47	28.49	43.48	29.64	46.48	31.19
UltraSAM	79.69	70.82	67.19	55.39	47.55	38.18	67.23	56.45	79.06	69.69	73.38	62.13

Fig. 7 presents the qualitative analysis. In cross-center training and testing scenarios, UltraSAM demonstrated segmentation boundaries that closely aligned with the GT, particularly excelling in capturing details in the regions with complex shapes or irregular edges. Certain methods, such as TransUNet and SegFormer, tended to produce over-segmentation or under-segmentation in low-contrast areas, leading to deviations from the GT. When trained on the Mix dataset, UltraSAM exhibited stable segmentation performance across various types of lesions, including circular, elliptical, and irregular shapes. In contrast, certain methods, such as SwinUNet and TransFuse, were prone to generating artifacts or discontinuities in detailed regions, resulting in incomplete segmentation outcomes. These results highlighted the robustness and accuracy of the proposed method in diverse segmentation challenges.

4.7. Comparison with SOTA foundation models on the external test set

Further evaluation was conducted to compare UltraSAM against SAM-based baseline models on cross-center ultrasound segmentation tasks. As shown in Table 6 and Fig. 6, the results demonstrated that when trained on the BUSI dataset, UltraSAM exhibited modestly lower performance than SAMUS in breast and thyroid segmentation tasks, suggesting that SAMUS might possess greater sensitivity in capturing features specific to breast nodules. However, when trained on the DDTI and Mix datasets, UltraSAM consistently outperformed all six baseline models. This performance advantage underscored the effectiveness of the parallel feature fusion architecture of UltraSAM in addressing domain shifts inherent in ultrasound imaging data. In summary, training UltraSAM on the Mix dataset enhanced its generalization capabilities across diverse tasks and centers by leveraging increased data diversity. This improvement established a robust foundation for its potential deployment in clinical multi-task scenarios; this will be further explored in subsequent sections.

Fig. 7 presents the qualitative analysis. Certain methods, such as SAM and SAMUS, performed well on certain datasets but exhibited performance fluctuations on others. This highlighted the superior

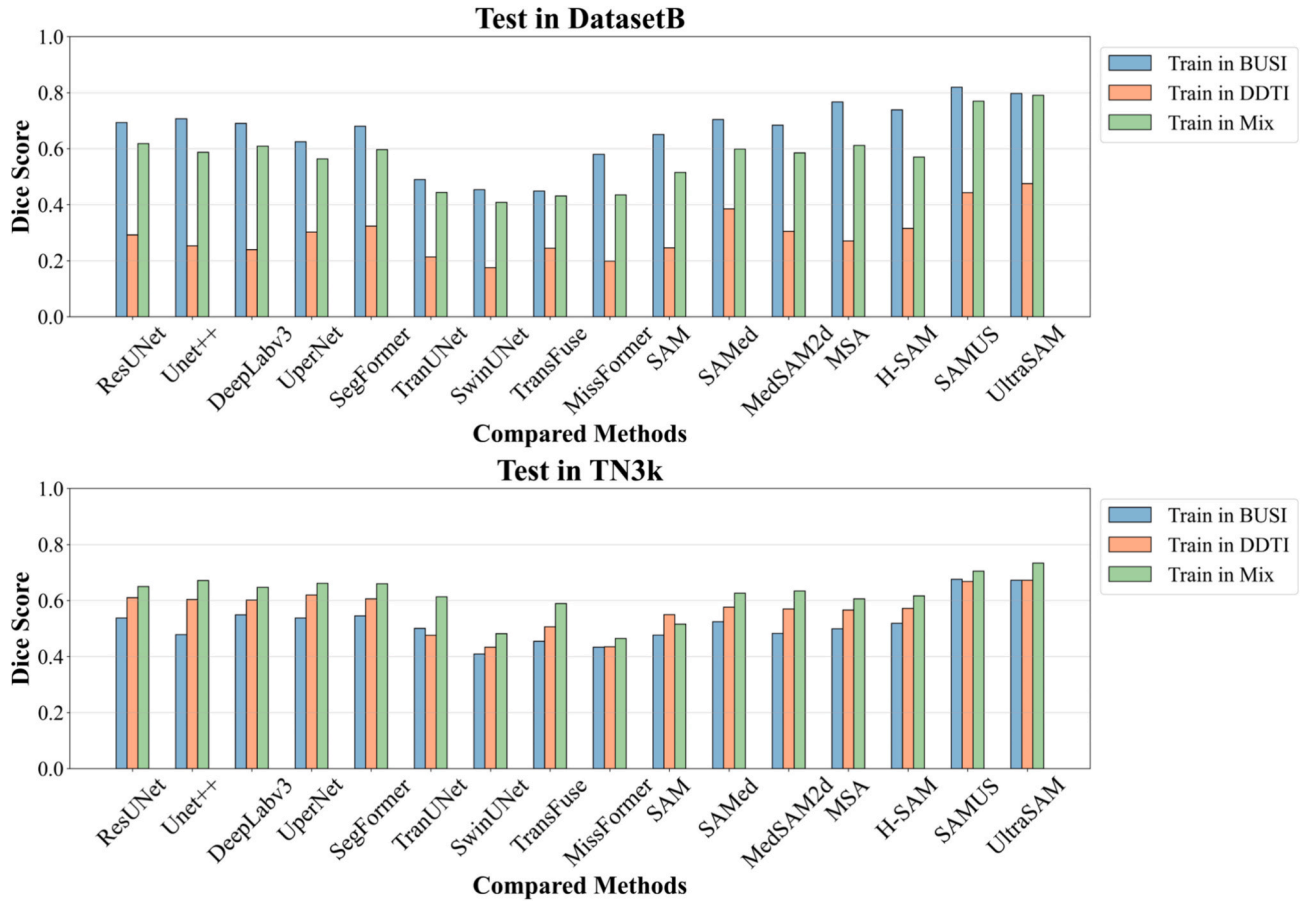


Fig. 6. Comparison between UltraSAM and task-specific and foundation methods evaluated on Dataset B (top) and the TN3k (bottom) external test set.

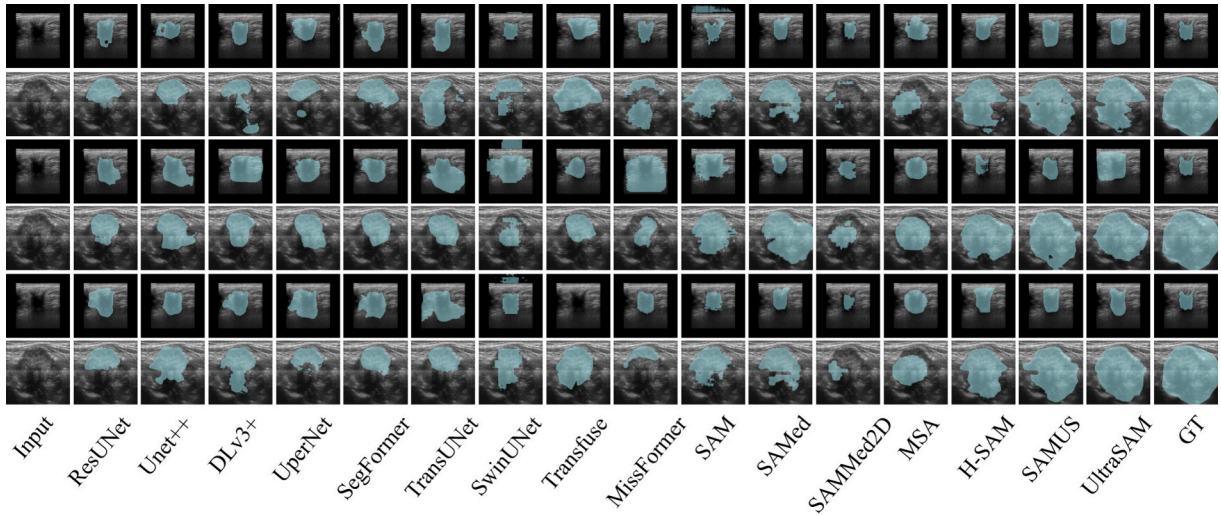


Fig. 7. Qualitative comparisons between UltraSAM and task-specific and foundation methods on the external test set. Row 1: training on BUSI and testing on Dataset B (BUSI-Dataset B); Row 2: BUSI-TN3k; Row 3: DDTI-Dataset B; Row 4: DDTI-TN3k; Row 5: Mix-Dataset B; Row 6: Mix-TN3k.

generalization capability of UltraSAM that maintained consistent performance across diverse datasets.

4.8. Comparison with SOTA task-specific methods on the zero-shot generalization test set

We systematically evaluated the performance of UltraSAM on six unseen anatomical structures—fetal brain, liver, kidney, heart, prostate,

and thyroid—using a model trained on the Mix dataset. The results were compared against those of task-specific segmentation methods to assess cross-organ generalization capabilities.

UltraSAM demonstrated significant advantages in cross-organ generalization. As presented in Table 7 and Fig. 8, it consistently outperformed nine baseline methods across all tasks based on both Dice and IoU metrics. Notably, UltraSAM achieved an average Dice score of 73.23 %, representing a 13.19 % improvement over the second-best method,

Table 6

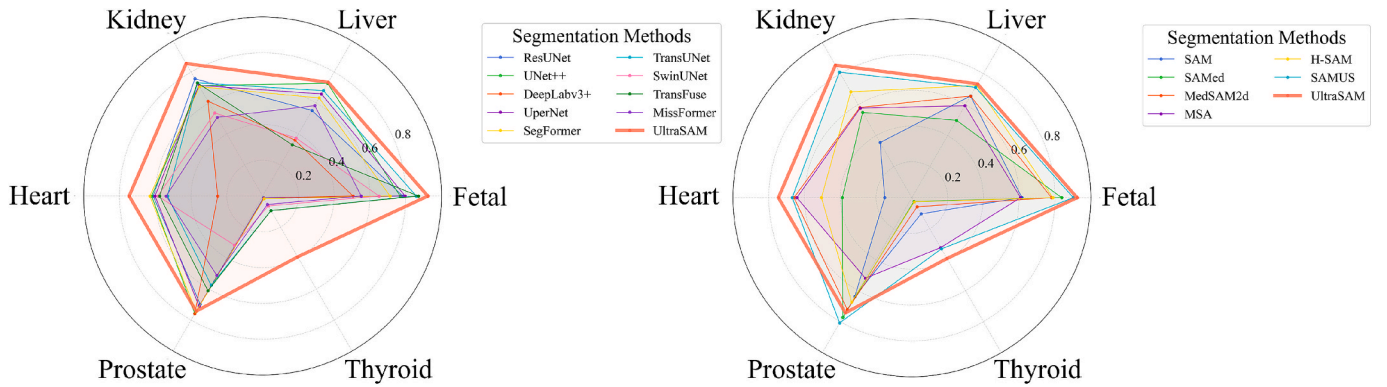
Quantitative comparison of UltraSAM and foundation models on the external test set. The best results are marked in bold.

Training set	BUSI				DDTI				Mix			
Test set	DatasetB		TN3k		DatasetB		TN3k		DatasetB		TN3k	
Methods	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
SAM	65.04	53.85	47.65	40.17	24.58	23.83	54.99	41.35	51.51	40.74	51.56	38.33
SAMed	70.37	61.23	52.42	42.54	38.51	31.33	57.61	46.65	59.85	52.48	62.62	52.21
SAM-Med2D	68.35	54.12	48.25	76.10	30.52	20.35	57.01	43.21	58.47	50.22	63.40	57.52
MSA	76.65	68.04	49.92	34.87	27.05	17.83	56.64	42.88	61.13	60.96	60.56	55.91
H-SAM	73.86	65.94	51.85	41.98	31.49	26.57	57.18	46.47	56.97	48.67	61.69	51.35
SAMUS	81.92	73.09	67.56	55.61	44.27	34.28	66.80	55.83	76.98	67.76	70.47	59.25
UltraSAM	79.69	70.82	67.19	55.39	47.55	38.18	67.23	56.45	79.06	69.69	73.38	62.13

Table 7

Quantitative comparison of UltraSAM and SOTA task-specific methods on the zero-shot generalization test set. The best results are marked in bold.

Test set	Fetal		Liver		Kidney		Heart		Prostate		Thyroid	
Methods	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
ResUNet	78.39	65.82	55.18	39.31	75.65	61.42	61.19	45.11	69.79	55.33	01.09	00.59
UNet++	76.86	64.46	72.70	58.05	71.19	55.81	62.06	45.88	75.81	62.22	01.62	00.86
DeepLabv3+	50.54	35.29	36.10	23.61	61.22	44.98	25.09	14.97	75.92	62.30	01.30	00.69
UperNet	79.71	67.16	65.83	50.53	71.39	55.90	60.16	43.69	71.07	56.53	01.36	00.71
SegFormer	71.24	57.53	63.27	48.11	70.54	55.00	63.06	46.80	74.44	60.76	01.65	00.88
TransUNet	85.92	75.70	67.96	52.40	72.99	57.94	53.20	36.86	57.59	41.86	09.41	05.80
SwinUNet	65.13	48.73	37.43	23.56	53.54	36.88	58.62	42.62	31.65	19.69	06.42	03.69
TransFuse	87.06	77.45	33.09	21.22	73.00	57.85	57.64	41.80	61.12	45.69	09.37	05.82
MissFormer	55.15	38.85	58.34	41.86	50.75	34.46	53.92	37.68	51.23	35.34	05.41	03.10
UltraSAM	92.35	86.41	73.39	59.52	85.47	76.57	74.60	60.90	74.31	62.80	39.24	28.59

**Fig. 8.** Comparison of UltraSAM versus task-specific methods (left) and foundation methods (right) in the zero-shot generalization test set.

UNet++ (60.04 %). In fetal brain segmentation, UltraSAM attained exceptional Dice and IoU scores of 92.35 % and 86.41 %, respectively, highlighting the benefits of diverse training data for feature generalization. CNN methods exhibited marked performance variability in complex organ segmentation, whereas Transformer-based models frequently suffered from local detail loss in cross-domain tasks. In contrast, UltraSAM maintained robust performance across anatomically distinct structures. For instance, it achieved Dice scores of 85.48 % for kidneys and 39.24 % for thyroid glands—both superior to those of all baseline methods. Despite lower thyroid segmentation scores attributable to morphological differences between training and test datasets, UltraSAM significantly outperformed alternative approaches that achieved Dice scores ranging from 1.09 % to 9.41 %. In heart segmentation, SegFormer achieved a Dice score of 63.06 %, demonstrating the utility of multi-scale feature fusion and lightweight decoders for chamber structure modeling. However, UltraSAM further improved performance to 74.61 % through its innovative frequency domain feature integration. These results collectively confirmed that the hybrid architecture of UltraSAM—combining cross-organ data augmentation and hierarchical feature interaction—substantially enhanced its ability to generalize to

previously unseen anatomical structures. This capability represents a significant advancement in addressing the challenges of multi-organ segmentation across diverse ultrasound imaging modalities.

Fig. 9 presents the qualitative analysis. Compared with task-specific methods, the proposed method demonstrated superior performance in capturing segmentation boundaries that closely align with the GT, particularly in regions with complex shapes or irregular edges, such as liver and kidney segmentation. Other methods often exhibited blurred boundaries or structural omissions in these challenging areas. In handling complex structures (for example, heart and prostate regions), the proposed method achieved complete segmentation of the target area without noticeable under-segmentation or over-segmentation. In contrast, methods such as TransUNet and SegFormer were prone to over-segmentation or under-segmentation in these scenarios. For fine structural details (for example, small regions in kidneys and thyroid), the proposed method effectively preserved the target details, avoiding erroneous merging or omission. Conversely, methods such as SwinUNet and TransFuse tended to generate artifacts or discontinuities in detailed regions, resulting in incomplete segmentation outcomes. These results underscored the robustness and accuracy of the proposed method in

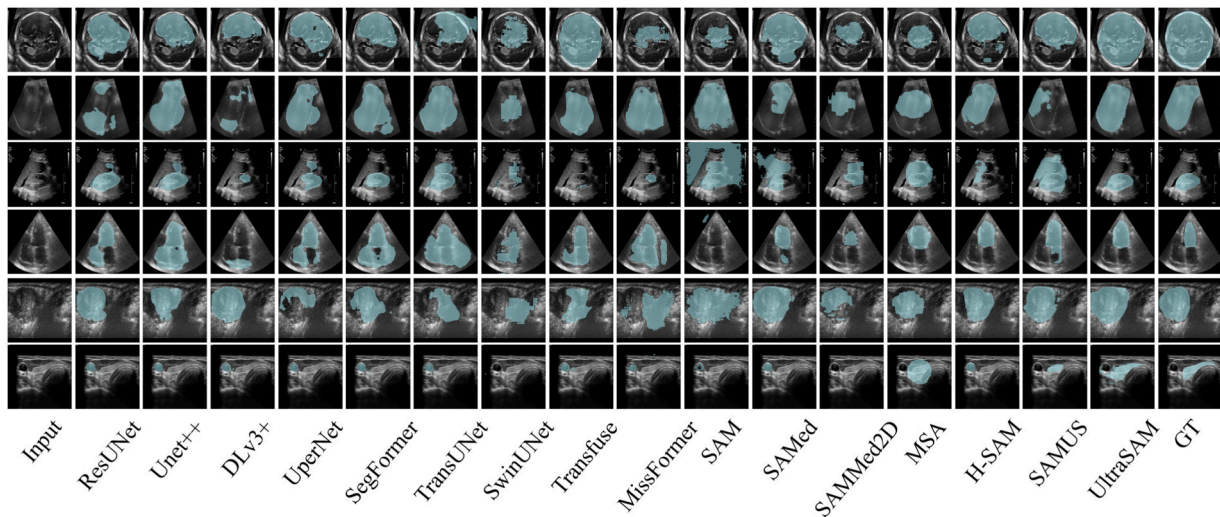


Fig. 9. Qualitative comparisons between UltraSAM and task-specific and foundation methods on the zero-shot generalization test set. Row 1: training on Mix and testing on Fetal (Mix-Fetal); Row 2: Mix-Liver; Row 3: Mix-Kidney; Row 4: Mix-Heart; Row 5: Mix-Prostate; Row 6: Mix-Thyroid.

addressing diverse segmentation challenges.

4.9. Comparison with SOTA foundation models on the Zero-Shot generalization test set

Table 8 and Fig. 8 presents a comparative analysis of various SAM-based methods across six unseen anatomical structures. The results demonstrated that UltraSAM achieved superior performance relative to other SAM variants, attaining an average Dice score of 76.70 %; this represented the highest score among all evaluated methods. UltraSAM demonstrated exceptional performance in fetal segmentation, achieving a Dice score of 92.35 %, surpassing that of SAMUS by 2.07 % and SAMed by 8.60 %. Notably, in thyroid gland segmentation—a task characterized by significant morphological discrepancies between training and test datasets—UltraSAM achieved the highest Dice score of 39.24 % among all methods, despite a lower performance relative to other organs. Other SAM variants exhibited varying degrees of specialization and generalization limitations. SAMed achieved strong performance in fetal brain (Dice: 83.76 %) and prostate (Dice: 77.18 %) segmentation but demonstrated severe generalization failure in thyroid (Dice: 2.39 %) and heart (Dice: 38.92 %) segmentation. H-SAM exhibited robust performance in liver (Dice: 78.46 %) and kidney (Dice: 68.27 %) segmentation but underperformed in heart segmentation (Dice: 50.55 %). SAMUS achieved optimal prostate segmentation (Dice: 80.79 %) through multi-branch joint training but lagged behind UltraSAM in segmentation performance for other organs.

Overall, UltraSAM demonstrated the most balanced generalization performance across unseen organs by integrating cross-organ data augmentation and frequency domain feature fusion mechanisms. Its standard deviation of Dice scores was significantly lower than those of SAM series models and other comparison methods, validating the

robustness of UltraSAM in cross-domain medical image segmentation.

Fig. 9 presents the qualitative analysis. UltraSAM demonstrated stable segmentation performance across various organs. H-SAM and SAMUS showed reduced adaptability across different organ types and were unable to simultaneously handle multiple structures. The proposed method significantly enhanced segmentation accuracy and robustness through its innovative network design. It surpassed other foundational models in complex boundary handling, detail preservation, and cross-organ generalization capabilities. We further analyzed how domain differences across organs affect the transferability of RGB pretraining. In the fetal head and kidney tasks, where geometric boundaries are clear and structures are stable, the general edge priors from RGB pretraining yield pronounced positive transfer when combined with frequency-domain injection. In contrast, the thyroid task features low boundary contrast and substantial morphological divergence of nodules relative to the training domain, increasing the risk of negative transfer. Although UltraSAM outperforms other methods, its overall score remains lower than in other organs, indicating a need to further expand cross-organ training diversity.

We benchmarked the training parameter count of UltraSAM against existing SAM variants. UltraSAM has only 17.37 M trainable parameters, ranking second among all SAM variants and amounting to approximately 2.7 % of SAM (636.01 M) and 43 % of SAMUS (39.95 M). These results confirm that UltraSAM attains more accurate segmentation and stronger generalization at lower computational expense.

4.10. Comparison with pretrained SOTA foundation models

We evaluated the generalization capabilities of the SAM, SAMUS, and UltraSAM without fine-tuning. The SAM was pre-trained on more than one billion natural images and SAMUS on a diverse ultrasound

Table 8

Quantitative comparison of UltraSAM and foundation models on the zero-shot generalization test set. The best results are marked in bold.

Test set	Fetal		Liver		Kidney		Heart		Prostate		Thyroid		Training Parameter/M
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	
SAM	61.55	46.03	65.58	49.91	35.55	23.78	15.15	9.25	63.89	48.62	10.42	6.48	636.01
SAMed	83.75	74.38	49.89	38.65	55.01	44.85	38.91	30.31	77.17	66.42	2.38	1.41	18.81
SAM-Med2D	78.10	70.62	65.59	49.94	58.20	44.90	65.89	50.11	72.19	61.91	5.86	3.29	184.57
MSA	60.78	44.83	59.33	43.19	57.74	42.02	64.29	47.95	51.97	36.16	32.20	25.23	13.02
H-SAM	78.85	70.69	72.46	67.15	68.26	58.53	50.54	40.10	67.35	54.71	2.71	1.70	20.70
SAMUS	90.28	83.03	71.16	58.04	81.00	70.95	66.89	52.19	80.78	70.77	33.03	22.62	39.95
UltraSAM	92.35	86.41	73.39	59.52	85.47	76.57	74.60	60.90	74.31	62.80	39.24	28.59	17.37

dataset containing six organ types with 30,000 + annotated images, whereas UltraSAM was trained using only two types of ultrasound images with approximately 600 annotations.

As presented in Table 9, both SAMUS and UltraSAM significantly outperformed the SAM across six organ segmentation tasks, demonstrating the limited applicability of the SAM to medical segmentation owing to the domain gap between natural and medical images. UltraSAM achieved superior performance than SAMUS in three unseen organ segmentation tasks (fetal, liver, and kidney). Conversely, SAMUS showed better results in heart, prostate, and thyroid segmentation. Notably, the training data of SAMUS included heart and thyroid cases, explaining its strong performance on these specific tasks. For prostate segmentation, UltraSAM maintained competitive performance with only a 3.228 % reduction in the Dice score compared to the Dice score of SAMUS, despite using approximately 2 % of the training data of SAMUS. These results confirmed that UltraSAM achieved exceptional generalization performance with extremely limited training data while maintaining domain-specific adaptability.

5. Discussion

The SAM exhibits suboptimal performance when applied to ultrasound images. Additionally, SAM necessitates a large amount of annotated data for effective training, which can be a significant limitation in practical applications. Consequently, we must address a more clinically relevant problem: achieving robust cross-domain generalization with only a few samples. This new objective demands a fundamentally more sophisticated architecture than merely concatenating existing modules. To address these limitations of SAM, we introduce UltraSAM, an enhanced version of SAM that incorporates three novel modules: the FFTConv branch, the FFTI, and the LoRA module. These modules are specifically designed to improve the model's performance on ultrasound images and reduce its dependency on extensive annotated data.

Ultrasound image formation relies on reconstructing echo signals through Fourier transform in which frequency-domain information directly reflects tissue acoustic impedance differences (Hasegawa, 2021). Traditional CNNs are highly susceptible to speckle noise when extracting spatial-domain features (Ilesanmi & Ilesanmi, 2021), whereas FFT Conv branch leverages frequency-domain analysis to effectively separate noise from anatomical structures. However, the global attention mechanism of ViT exhibits reduced sensitivity to local high-frequency details (Azad et al., 2023), frequently resulting in loss of fine boundaries for small targets such as micro-nodules. Although FFT has been previously explored in generic SAM adapters (Wei et al., 2024), UltraSAM is the first to deeply couple FFT with the physical characteristics of ultrasound imaging; through the synergistic design of FFTConv–FFTI–LoRA, it achieves robust ultrasound segmentation under small-sample, cross-modal, and multi-center conditions.

UltraSAM's FFTConv branch processes images in the frequency domain, using properties of ultrasound waves to improve feature extraction. Speckle noise forms clear patterns in frequency data, which helps separate it from real structures. Unlike spatial models, FFTConv identifies noise more effectively, increasing robustness (Table 2). Small, low-contrast lesions (e.g., early tumors) cause slight frequency changes. FFTConv detects these subtle shifts, complementing the ViT's global view and improving segmentation on BUSI and DDTI (Table 3). Performance often drops when using different ultrasound devices due to

varying frequency responses. FFTConv reduces this gap by learning frequency-based features, improving cross-device performance as shown in multi-center tests (Tables 5 and 6). Given the heterogeneous feature spaces of ViT and CNN branches, direct feature concatenation or addition would induce feature conflicts (Chen et al., 2022).

This study introduced the FFTI that dynamically allocated weights to ViT and FFT–CNN features at each encoder layer through cross-attention mechanisms. The FFTI learnt modality complementarity by adaptively emphasizing CNN features in speckle-prone regions and ViT features in homogeneous tissue regions, thereby achieving context-aware feature fusion. Crucially, this adaptive fusion mechanism directly addresses the fundamental domain gap between SAM's RGB-based pre-training and ultrasound physics. The FFTConv branch provides a physically-grounded, ultrasound-native feature stream. The FFTI then acts as a cross-modal gate, allowing the model to critically evaluate and selectively leverage SAM's spatial priors. SAM's knowledge of edge and structure is utilized where it aligns with frequency-domain evidence (e.g., for clear organ boundaries), but its potentially misleading biases from the natural image domain are overridden in regions dominated by ultrasound-specific artifacts like speckle. This strategic integration is key to achieving positive knowledge transfer, as evidenced by our strong cross-domain results (Tables 6–9).

To address the challenges associated with the extensive parameter requirements and high computational demands of fine-tuning the SAM, this study employ LoRA to fine-tune the ViT encoder of UltraSAM. This approach allows for the adjustment of weight matrices by training only a small fraction of parameters, which not only mitigates SAM's over-reliance on initial parameters but also suppresses overfitting. Compared with full fine-tuning, LoRA allows selective adaptation of the ViT encoder to ultrasound-specific features while preserving SAM's general edge and shape detection capabilities. This ensures that only beneficial aspects of the RGB pretraining are retained. As a result, UltraSAM achieves robust domain adaptation for ultrasound imaging tasks even with limited training data and computational resources.

The performance of UltraSAM was evaluated across four distinct tasks, yielding promising results. When trained on the Mix dataset, UltraSAM achieved superior segmentation performance for thyroid and breast on homologous test sets, attaining a Dice score of 79.85 % on the BUSI breast segmentation task and 85.15 % on the DDTI thyroid segmentation task. In cross-center testing scenarios, UltraSAM achieved Dice scores of 79.06 % on the Dataset B breast segmentation task and 73.38 % on the TN3k thyroid segmentation task. Furthermore, generalization experiments across six different organ segmentation tasks demonstrated that the models trained using UltraSAM on small datasets exhibited strong domain portability. Existing foundational medical SAM variants such as MedSAM and SAM-Med2D have achieved remarkable success in segmentation tasks for CT and MRI. However, for ultrasound imaging scenarios—which are characterized by heightened noise and distinctive frequency-domain attributes—methods employing frequency-domain feature enhancement may offer more straightforward and effective segmentation performance. As demonstrated in our experiments (Tables 4, 6 and 8), UltraSAM consistently outperforms these models on multiple US datasets. These results underscore the necessity of a tailored architecture, such as our FFT-enhanced design, rather than direct application of generic medical SAM extensions.

Some studies have pointed out that Datasets may restrict to single organs, such as those containing only thyroid or breast images, lack

Table 9

Quantitative comparison of UltraSAM and pretrained foundation models on the zero-shot generalization test set. The best results are marked in bold.

Test set	Fetal		Liver		Kidney		Heart		Prostate		Thyroid	
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
SAM	19.01	13.61	20.10	15.09	18.02	15.87	28.97	26.57	47.25	36.81	11.09	6.50
SAMUS	89.60	82.82	73.20	59.31	77.81	66.68	83.37	73.84	77.53	66.56	89.86	82.14
UltraSAM	92.35	86.41	73.39	59.52	85.47	76.57	74.60	60.90	74.31	62.80	39.24	28.59

sufficient anatomical diversity to support generalization across other organs (Fang & Yan, 2020). To address this limitation, this study introduced an innovative approach by employing a mixed segmentation task of thyroid and breast nodules as the training set. This strategy was motivated by the observation that thyroid and breast nodules exhibited both similarities and complementary characteristics in ultrasound imaging.

As shown in Fig. 10, thyroid ultrasound images typically contained nodules with simpler morphologies, whereas breast ultrasound images often included nodules with more complex structures. Additionally, hypoechoic nodules were less frequent in breast images but more prevalent in thyroid images. These distinct differences in the nodule shape and boundary characteristics challenged the model to learn more discriminative features, thereby enhancing its ability to extract generalizable representations of ultrasound images. Through data augmentation and transfer learning, even a limited number of samples can effectively cover common variations in ultrasound imaging, thereby mitigating overfitting risks. Experimental results demonstrated that the mixed dataset training strategy not only preserved single-organ segmentation accuracy but also significantly improved the cross-organ and cross-center generalization capabilities of the model, with a dice improvement of 0.32 % and 6.15 % in internal, external test set for the segmentation of thyroid nodules. These findings validated the potential of the strategy for application in clinical multi-task scenarios.

Real-time inference and strong generalization capabilities of UltraSAM make it a promising candidate for integration into portable medical devices, enabling intraoperative real-time segmentation of organs and lesions. Although UltraSAM, like most SAM-based models, requires a

prompt to specify the target, the associated clinical overhead is minimal, and is deemed acceptable within diagnostic routines. Furthermore, deploying UltraSAM in primary care hospitals can significantly enhance medical annotation efficiency of assisting physicians in annotating diverse data types. These applications highlight the potential of UltraSAM to bridge the gap between advanced artificial intelligence capabilities and practical clinical requirements, supporting both technological integration and improved diagnostic workflows in resource-limited settings.

UltraSAM's real-time inference and strong generalization capabilities make it a promising candidate for integration into portable medical devices, enabling intraoperative real-time segmentation of organs and lesions. Clinical ultrasound diagnosis is a holistic process integrating segmentation, texture analysis, kinematic assessment, and clinician experience; the goal of UltraSAM is not to replace this comprehensive analysis but to provide a reliable and automated tool for the initial, often labor-intensive, step of delineating anatomical structures and lesions. By providing accurate and consistent segmentations, UltraSAM can standardize quantitative measurements and reduce inter-observer variability, establishing a solid foundation for downstream tasks such as disease classification, severity staging, and treatment planning, and ultimately supporting the clinician's expert judgment. Although UltraSAM, like most SAM-based models, requires a prompt to specify the target, the associated clinical overhead is minimal and deemed acceptable within diagnostic routines. Furthermore, deploying UltraSAM in primary care hospitals can significantly enhance medical annotation efficiency by assisting physicians in annotating diverse data types. These applications highlight the potential of UltraSAM to bridge the gap

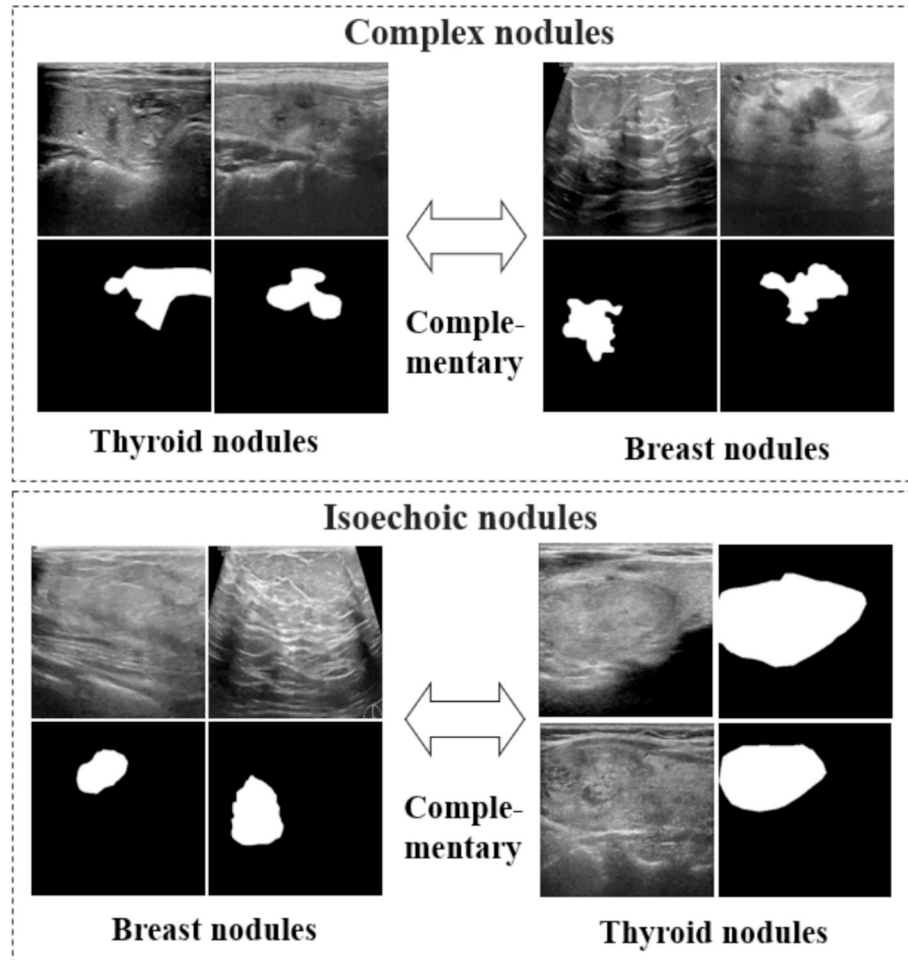


Fig. 10. Comparison of characteristics of breast and thyroid nodules in ultrasound images.

between advanced artificial intelligence capabilities and practical clinical requirements, supporting both technological integration and improved diagnostic workflows in resource-limited settings.

Although UltraSAM demonstrates significant potential, certain limitations warrant further investigation. First, UltraSAM current relies on manual input for prompts. For fully automated screening scenarios, future work may incorporate lightweight object detectors (e.g., the YOLO family) to generate prompts. Second, in scenarios where the target region exhibits high variability or low contrast with surrounding tissues, performance may be suboptimal, indicating areas for potential improvement. Third, although PEFT has reduced the number of training parameters, the inference-time parameter size of the model remains relatively large. Future research will focus on developing lightweight architectures and automated prompt generation to address these limitations while maintaining performance standards.

6. Conclusion

In summary, this study introduced UltraSAM, a deep learning-based foundational segmentation model specifically designed for ultrasound imaging applications. UltraSAM could accurately segment diverse anatomical structures and pathological features despite being trained on a relatively limited dataset comprising hundreds of images. Rigorous internal and external validation demonstrated that UltraSAM exhibited superior generalization capabilities compared to existing state-of-the-art task-specific methods and SAM-based foundation methods, even when trained on limited data. In zero-shot ultrasound segmentation, UltraSAM yields its largest gain (+59.45 %) on noisy, low-contrast cardiac images and its smallest gain (+2.07 %) on already high-performing fetal head images. These results highlighted its potential as a robust foundation model for medical ultrasound segmentation tasks.

7. Declaration of Generative AI and AI-assisted technologies in the writing process

The authors did not use Generative AI and AI-assisted technologies during the writing process in this work.

CRediT authorship contribution statement

Tao Jiang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Yifang Li:** Formal analysis, Investigation, Visualization, Writing – review & editing. **Wenyu Xing:** Formal analysis, Visualization. **Ran Cao:** Investigation, Software, Validation. **Ming Yu:** Investigation, Validation. **Yunkai Zhu:** Data curation. **Yaqing Chen:** Data curation. **Boyi Li:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **Dean Ta:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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